BUSINFO 702 – INFORMATION MANAGEMENT
2025 Q3 -Project Proposal

Project Title

A Data-Driven POV on United Nations Sustainable Development Progress impacting World Happiness.

Project Group

Group number: Group 08

Group members and roles:

S. No.	Name and Student ID	Role
1	Aryan Panchal (464282810)	Data Collection and Preparation Lead (ELT Process)
2	Shraddha Malve (273860286)	Data Warehouse and Star Schema Design Lead
3	Gibson G (141429531)	SQLite and Analytics Lead
4	Yiyi Zhang (871813506)	Documentation and Business Perspective Lead

Data Source

Dataset 1

1. **Source name**: Sustainable Development Data (2000-2023)
2. **Source type**: CSV File (Kaggle Dataset)
3. **Source size**: 4141 rows * 21 columns
4. **Source description**: A comprehensive dataset comprising country-wise sustainable scores of 17 sustainable development goals defined by the United Nations member states. It contains data from 180 countries from years 2000 to 2023.
    a. **Categorical variables**
        i. country_code - A unique identifier
        ii. country (180 unique values) - Name of the country
        iii. year (2000 to 2023) – Respective year of the observation
    b. **Numerical variables**
        i. sdg_index: Overall SDG index score of the country
        ii. goal 1 to goal 17 scores
5. **Source link**: https://www.kaggle.com/datasets/sazidthe1/sustainable-development-report

Dataset 2

1. **Source name**: World Happiness Index by Reports (2013 – 2023)
2. **Source type**: CSV File (Kaggle Dataset)
3. **Source size**: 1671 rows * 4 columns (with NULL values)

4. **Source description**: This dataset contains country-level happiness rankings and happiness index from years 2013 to 2023.
   a. **Categorical Variables**
      i. Country – 167 unique country names
      ii. Year – Report year
   b. **Numerical Variables**
      i. Index – Happiness Index
      ii. Rank – Happiness Rank
5. **Source link**: https://www.kaggle.com/datasets/simonaasm/world-happiness-index-by-reports-2013-2023.

Note: *The Sustainable Development Data (2000-2023) dataset consists of 17 sustainable development goals defined by the United Nations, and they are socio-economic in nature. For e.g.: No Poverty, Zero Hunger, Good Health and Well-being, Quality Education, Gender Equality, etc., to name a few.*

Project Motivation

Every country would like to give its people a better life. Some focus on building stronger economies, others on improving health, education, or protecting the environment. However, it is not always true whether progress in these sectors truly will lead to greater happiness amongst people.

The World Happiness Index (2013–2023) dataset tells us how people feel about their lives, reflecting trust, freedom, social support, and economic stability. The Sustainable Development dataset shows how nations meet the UN's 17 Sustainable Development Goals from ending poverty and hunger to tackling climate change and reducing inequality.

By bringing these two perspectives together, we can tell a more complete story about human progress holistically. We can see where sustainability and happiness progress together and drift apart. For example, if equality or environmental care is neglected, a country might grow richer but remain low in happiness. Another is that improving healthcare or education has an immediate positive effect on well-being.

Our purpose is to uncover these patterns so that leaders, policymakers, communities, and people can learn what drives sustainable progress in a country with human happiness. We expect the integration of these datasets to reveal both positive and negative relationships between Sustainable Development progress and happiness. The findings will provide actionable insights for policymakers, helping identify which development areas deliver the greatest well-being impact.

Project Plan

Our group will integrate the World Happiness Index by Reports (2013–2023) dataset with the Sustainable Development Report dataset to analyse the relationship between sustainable goals and global well-being.

1. Data Collection & Preparation

- Download both datasets from Kaggle.
- Inspect, clean, and standardise country names and years for joining.
- Select overlapping years (2013–2023) for analysis.

2. Data Loading (ELT Process)

- Load the cleaned data in SQLite
- Apply transformations to create and store attribute hierarchies and compute derived measures needed for analysis.

3. Data modelling (Star Schema Design in SQLite)

- **Fact table**: Stores main measures from both datasets for each country and year combination (Happiness Score, Sustainable Development Goal Index)
- **Dimension tables:** Country dimension, Time dimension (year) and between 3 and 5 Sustainable Development Goal indicators.

4. SQLite for Analytics

- Writing SQLite queries for at least three targeted analytical research questions that are directly related to the selected datasets.
- Ensuring that questions align with the project theme of sustainability and equity.
- Include both temporal (yearly) and regional (country) perspectives.

5. Report the findings

- Interpret the query results in a business context
- Identify key insights and create a comprehensive group project report.